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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE : DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	E-commerce Sales Analytics Dashboard	
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Abstract

The **E-commerce Sales Analytics Dashboard** is a data science and business analytics project developed to analyze large volumes of e-commerce sales data and convert raw data into meaningful and actionable insights. With the rapid growth of online shopping, e-commerce platforms generate huge amounts of data related to customers, products, orders, sales, revenue, discounts, payments, and delivery. Analyzing this data manually can be time-consuming and difficult. Therefore, this project aims to provide an interactive dashboard that presents important business information in a simple and understandable visual format.

The proposed system collects and processes e-commerce datasets containing information such as **order ID, order date, customer details, product name, product category, quantity, sales amount, discount, profit, location, and payment method**. The collected data is cleaned and preprocessed by handling missing values, removing duplicate records, correcting data inconsistencies, and transforming the data into a suitable format for analysis. Exploratory Data Analysis (EDA) is then performed to identify patterns, trends, relationships, and unusual values within the dataset.

The dashboard provides several important Key Performance Indicators (KPIs), including **total sales, total profit, total orders, total quantity sold, average order value, and number of customers**. It also provides visualizations for **monthly and yearly sales trends, category-wise sales, product-wise performance, regional sales, customer purchasing behavior, and profit analysis**. Charts such as bar charts, line charts, pie charts, and tables are used to make the information easier to understand.

The project can also help identify **best-selling and low-performing products, highly profitable categories, peak sales periods, and regions with high customer demand**. These insights can support businesses in improving inventory management, planning marketing strategies, identifying customer preferences, and increasing overall profitability.

The system can be implemented using technologies such as **Python, Pandas, NumPy,**

Matplotlib, Seaborn, and Streamlit/Power BI. Python can be used for data cleaning and analysis, while visualization tools can be used to create an interactive and user-friendly dashboard. The proposed dashboard reduces the effort required for manual data analysis and provides a centralized platform for monitoring e-commerce business performance.

Overall, the **E-commerce Sales Analytics Dashboard** demonstrates the practical application of **Data Science, Data Analytics, Data Visualization, and Business Intelligence** in the e-commerce domain. The project helps transform raw sales data into useful information and supports organizations in making faster, better, and data-driven business decisions.

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Project Guide